GDOT Traffic Dataset

Exploratory Data Analysis

2017 - 2024 | Georgia Department of Transportation

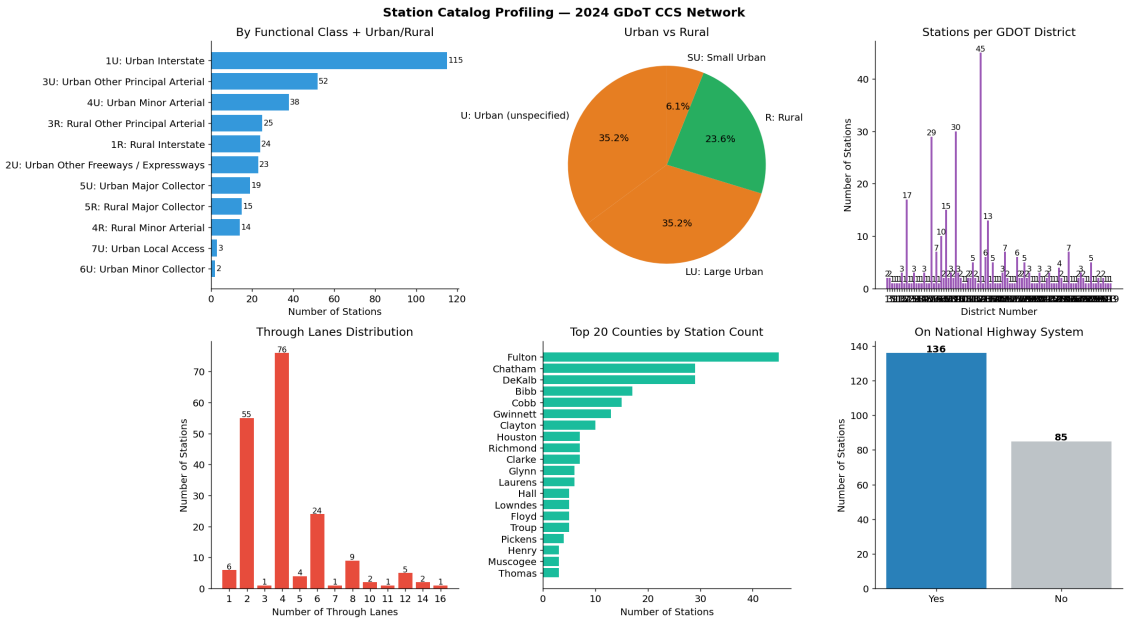
286 Stations * 3 Data Tiers * 7 Analysis Tracks

Executive Summary

Dataset at a Glance	
Stations in catalog	330 (286 with data)
Data tiers	3 (5-min 2024 / 5-min 2022-23 / Hourly 2017-22)
Coverage window	2017 - 2024 (8 years)
Station-months	21,837 total
Complete station-months	19,998 (91.6%)
Fleet missingness rate	8.4%
Functional classes	7 (FC1 Interstate dominant, 42%)
Urban / Rural split	~70% Urban, ~30% Rural

Data Trust & Modeling Readiness	
Trusted stations	243 (HIGH trust, >=2 sources)
CAUTION stations	36 (usable with caveats)
EXCLUDED stations	7 (flagged for investigation)
Trusted station-months	19,963 (96.9%)
Anomaly-days detected	391 (90.5% explained by event flags)
Reference station candidates	27 (all 5 quality criteria)
Clean hourly records (2024)	818,645 (99 stations)
Station feature table	286 stations x 27 features

Dataset Architecture & Schema Overview



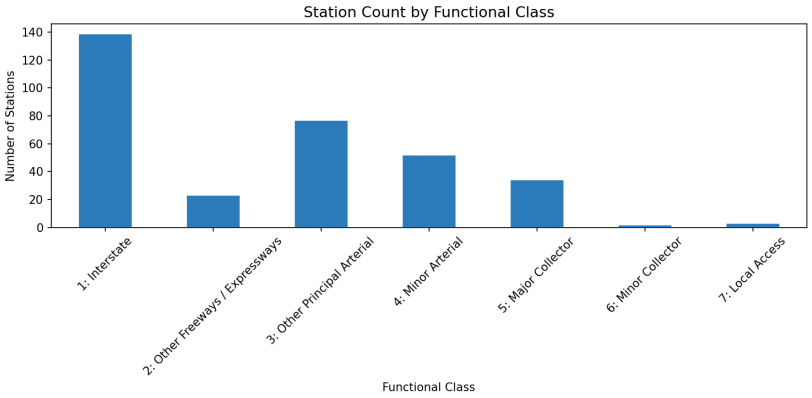
Three-Tier Architecture

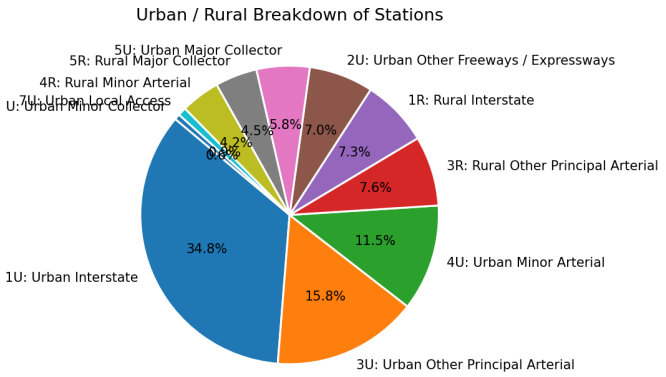
Tier 1 (2024)	5-min, directional, speed & weight bins
Tier 2 (2022-23)	5-min, lane-level, no speed/weight
Tier 3 (2017-22)	Hourly, directional, AvgSpeed + QC flags
All-3-tier overlap	225 stations (78.7%)
Exactly 2-tier	27 stations
Single-tier only	34 stations

Known Schema Issues

- Station ID: 12-digit COSIT (Tier 1) vs XXX-XXXX (Tiers 2-3) - requires lookup map
- 2 non-standard IDs (021-w334, 185-w227) prevent reliable cross-tier joins
- Tier 2 lacks speed/weight bins; Tier 3 is hourly not 5-min
- High metadata nulls (Bridge ID 99.7%, LTPP 98.8%) are field-type artefacts, not quality failures

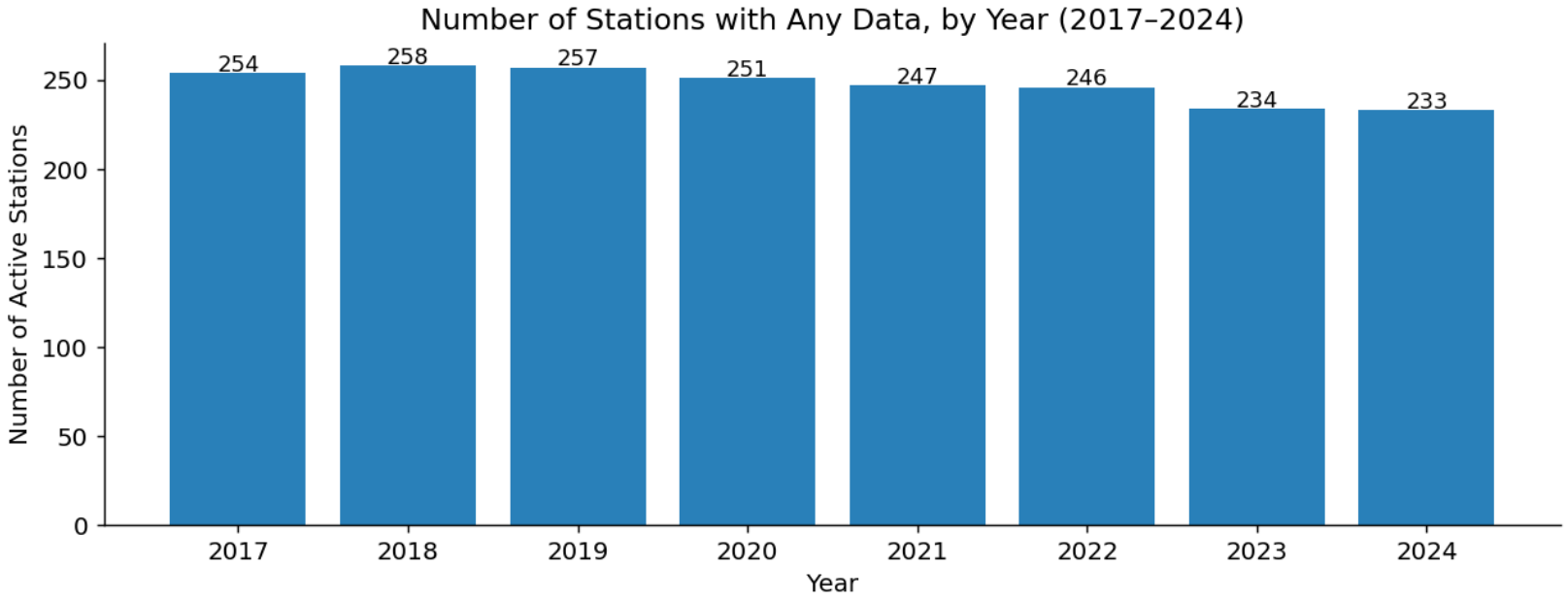
Data Inventory - Stations, Functional Class & Coverage





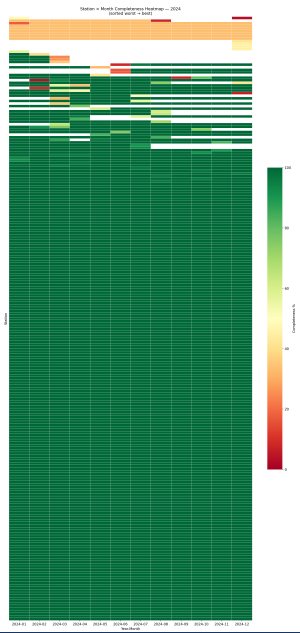
Key Numbers

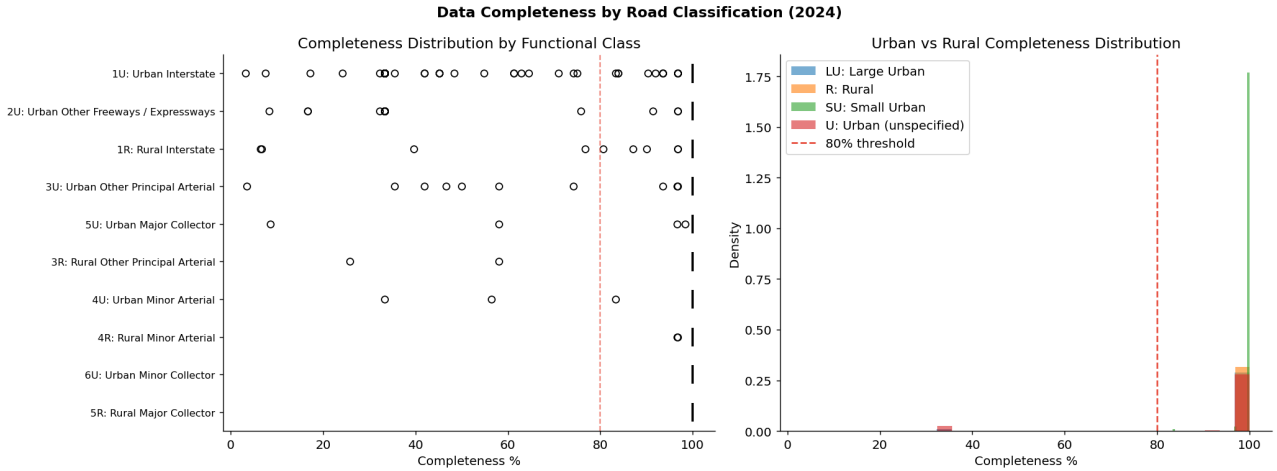
- FC1 Interstate: 139 stations (42%) - largest class
- FC3 Other Principal Arterial: 77 stations (23%)
- FC4 Minor Arterial: 52 * FC5 Major Collector: 34
- Urban Interstate leads urban split (115 stations)
- Active stations declining: 258 (2018) -> 233 (2024)
- Mean complete months per station: 72.9 (median 84.0)



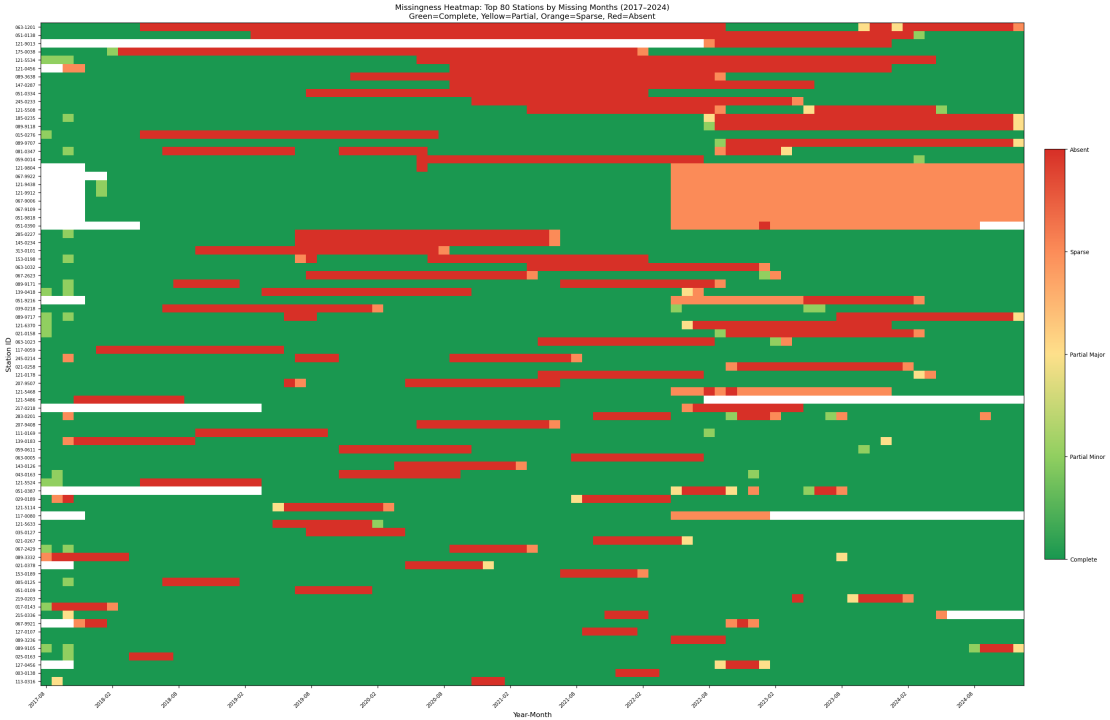
Completeness & Coverage by Tier and Station

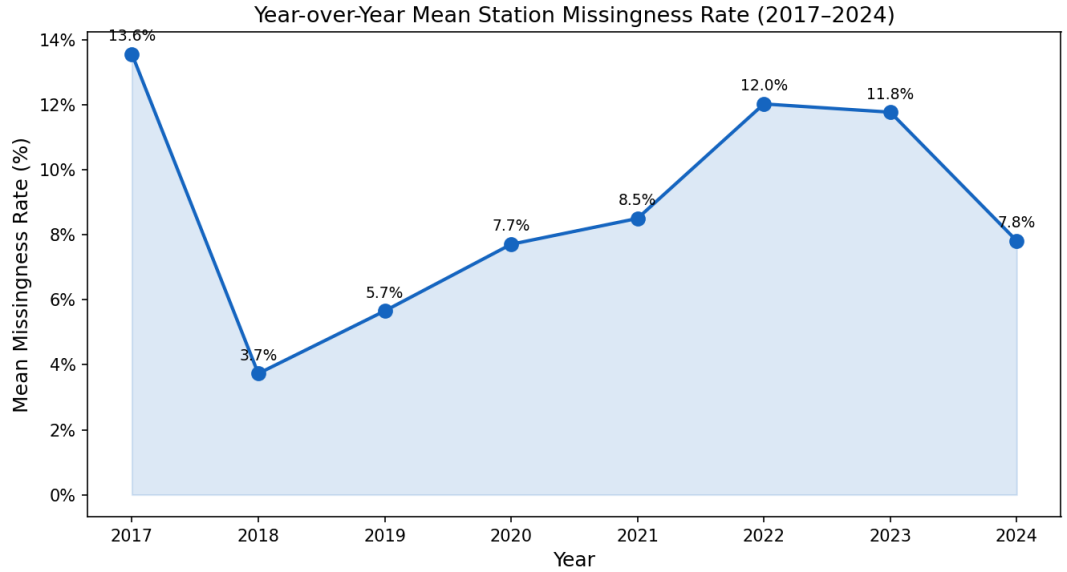






Missingness Audit - Overall & Temporal Patterns





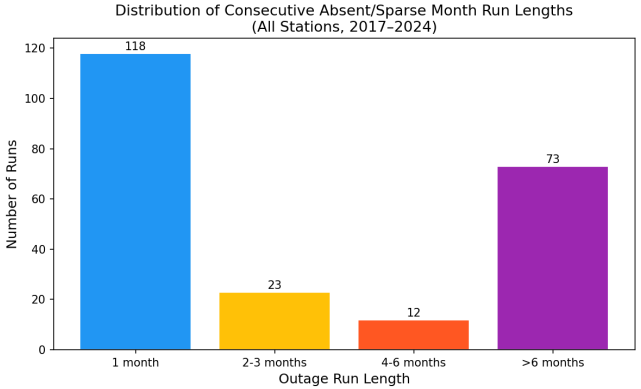
Fleet-Wide Findings

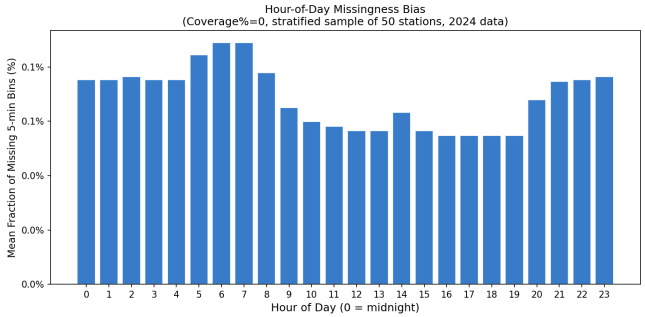
- Overall missingness: 8.4% of station-months below 80% coverage
- 73.4% of stations are 'always-on' (>=90% complete months)
- Only 1.8% are chronically sparse (<40% complete months)
- 226 outage runs identified; 73 runs exceed 6 months (max 60)

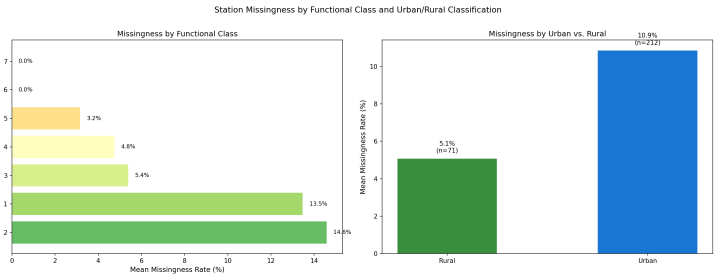
Year-over-Year Trend

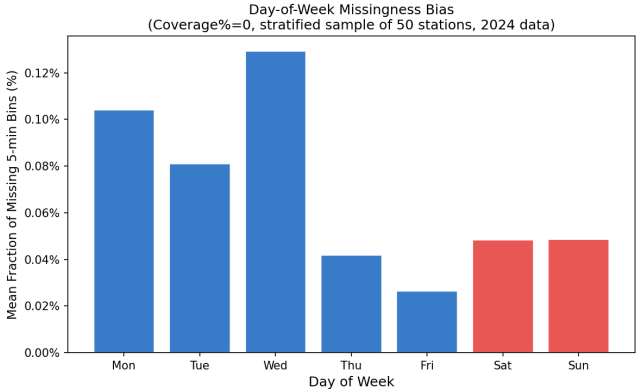
- 2017: 13.6% - incomplete initial deployment coverage
- 2022-23: 12% spike - system transition to 5-min pipeline
- 2024: 7.8% - partial recovery
- Pattern is operationally-driven, NOT random (not MCAR)

Missingness Patterns - Functional Class, Run Length & Hour Bias





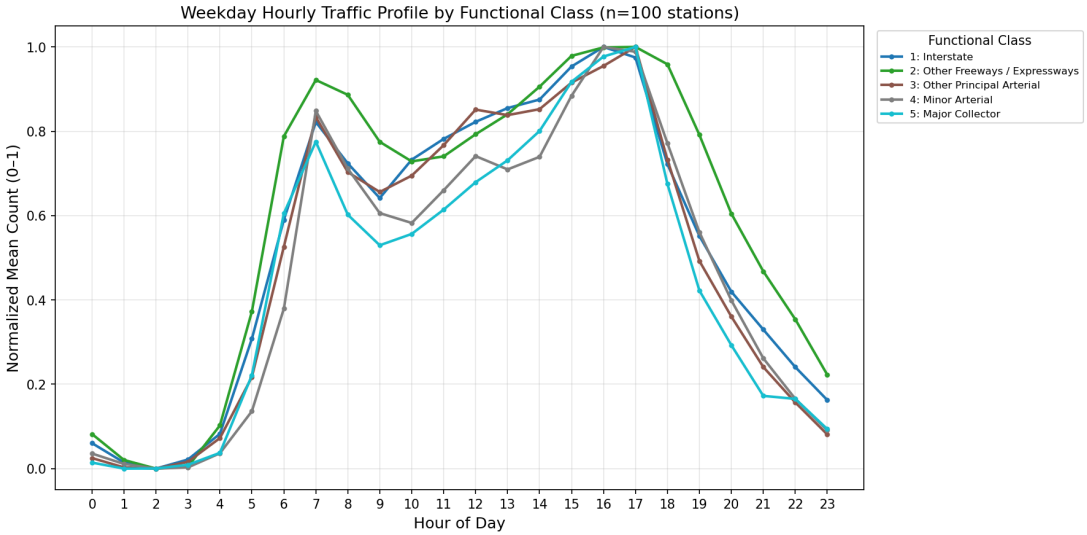


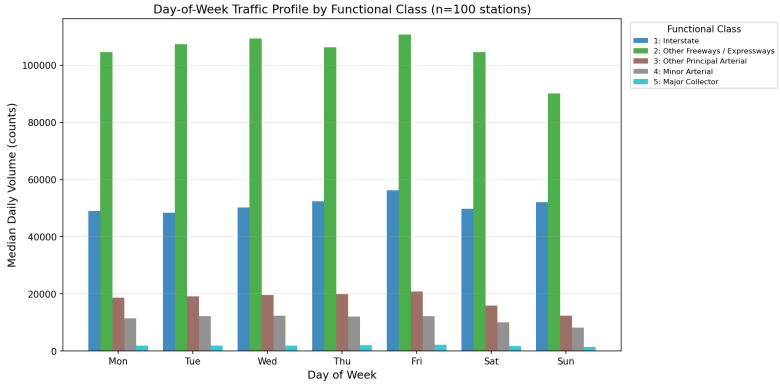


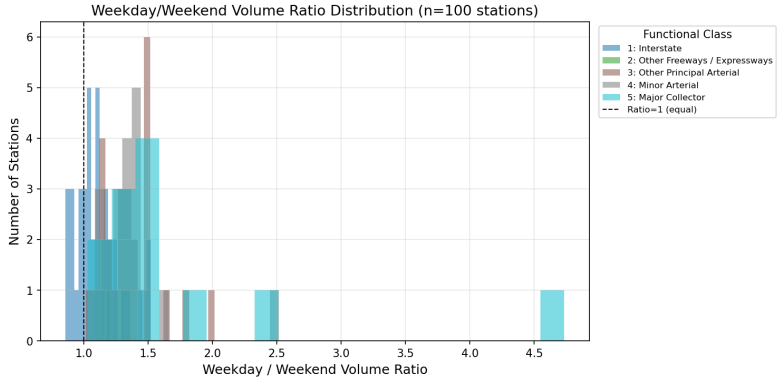
Key Pattern

- FC1 Interstate: 13.5% miss - highest class
- FC2 Freeways: 14.6% - likely multi-lane bin methodology
- FC3-5 Arterials/ Collector

Baseline Traffic Behavior - Profiles & Distributions







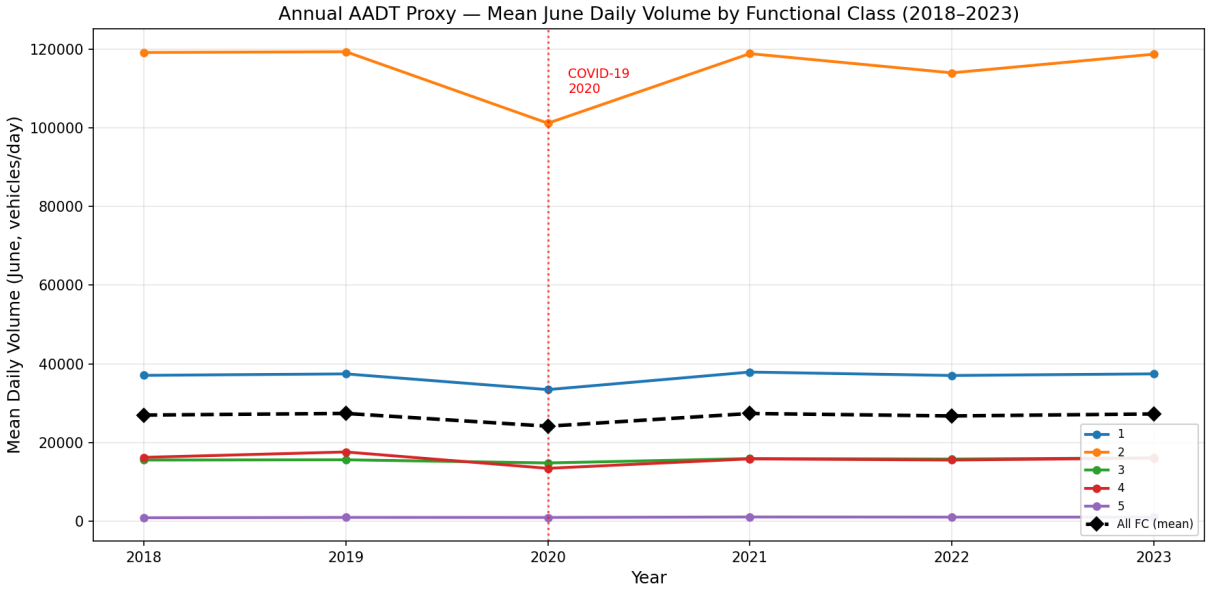
Volume Summary (100 stati

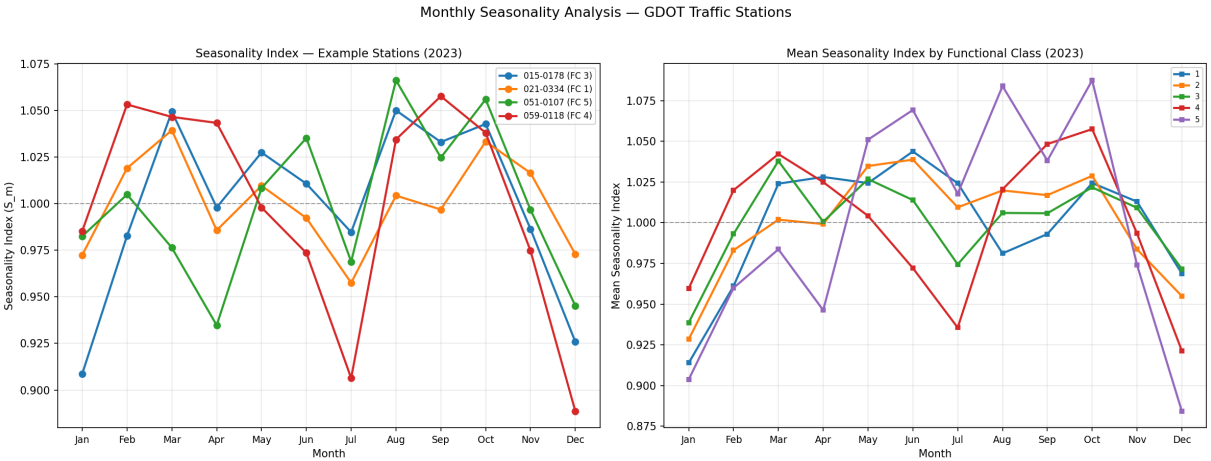
Median da	22,830 veh/da
Mean daily v	45,223 veh/da
FC2 Freew	~110K veh/da
FC1 Interstate	52K veh/da
FC3 OPA (	~19K veh/day
W/WE ratio (fleet)	1.05 (weekday
AM peak h	07:00 (stable)
PM peak hour	17:00 (stable)

Archetype Note

Station archetype clustering (k-means) collapsed to 77/78 stations 'Mixed' using 2024-only features. Functional class used as primary stratifier; archetype modeling deferred pending multi-year directional data.

Temporal Trends - AADT, Seasonality & Decomposition





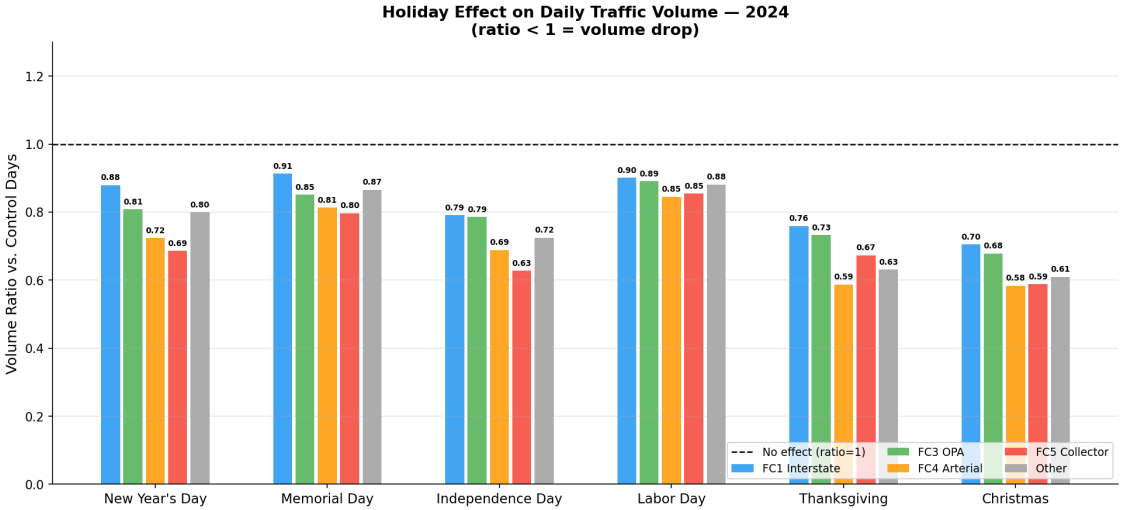
AADT Trends (30-station sample, 2018-2023)

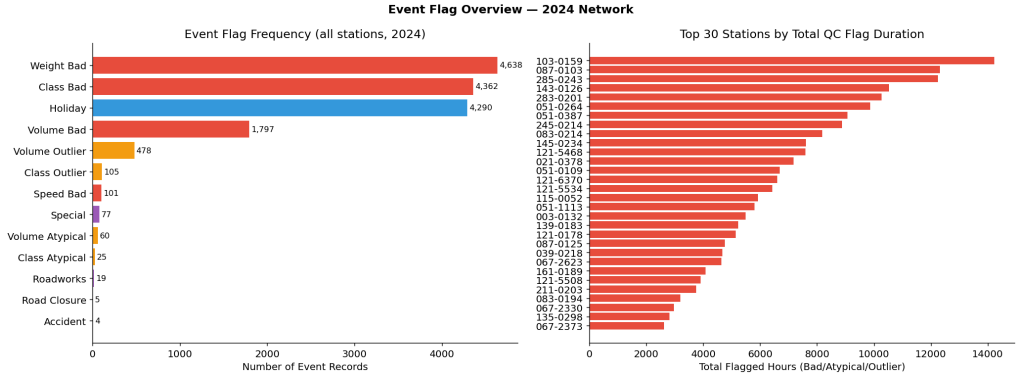
- COVID-19 impact: ~8.6% volume drop (2019->2020) across all FC
- Recovery visible by 2021-22; FC2 freeways recovered fastest
- AADT proxy = June mean daily volume (June is least seasonally biased)

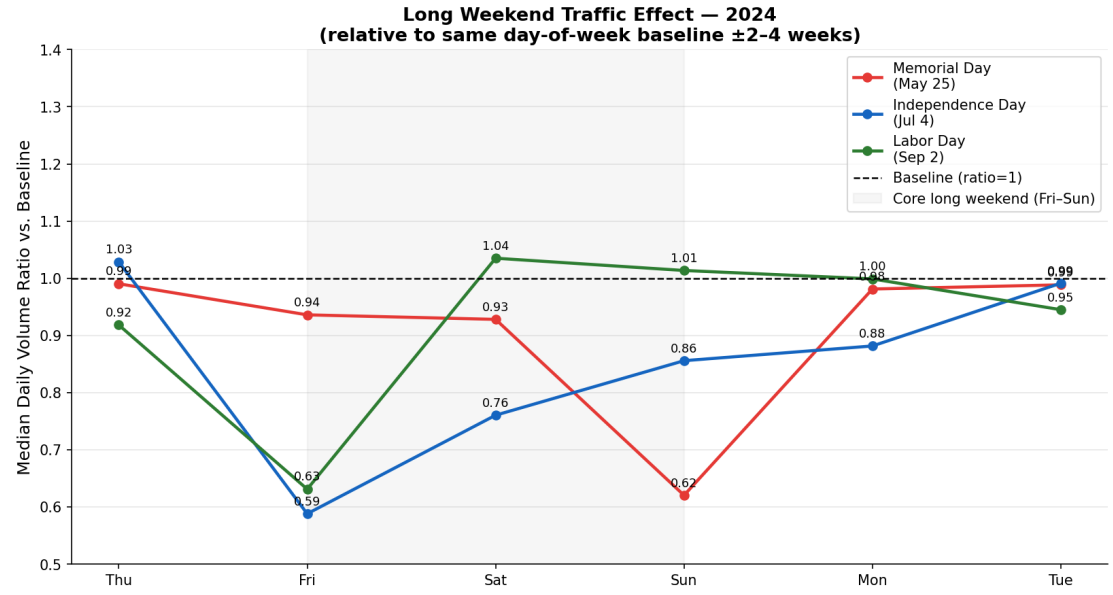
Seasonality (2023 baseline)

- Peak month: October (Seasonal Index = 1.036)
- Trough month: January (SI = 0.928) - +/-10% annual amplitude
- Rural FC4-5 stations show stronger seasonality than Interstate
- Peak-hour structure (AM 7:00, PM 17:00) stable across all 4 seasons

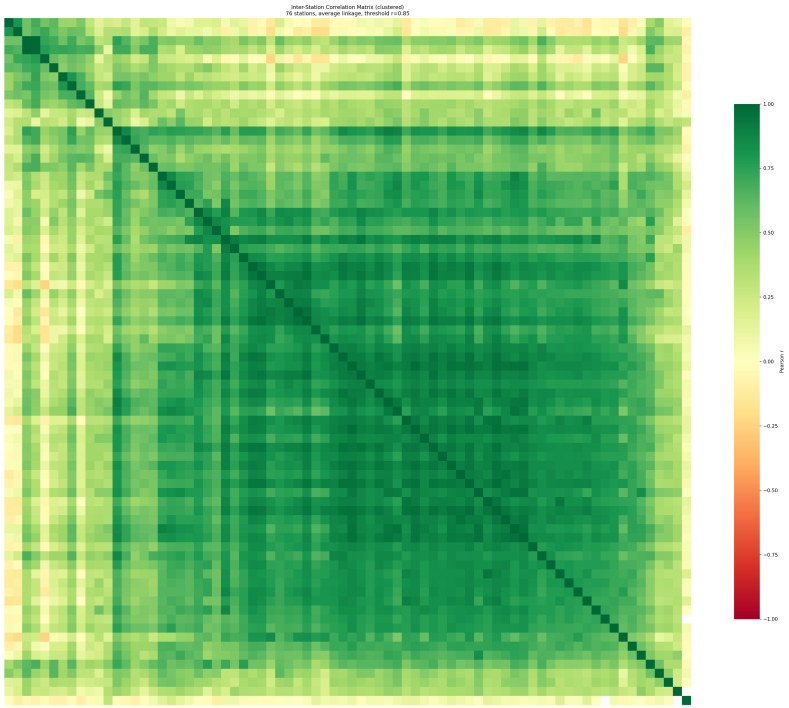
Holiday & Long Weekend Traffic Effects - 2024

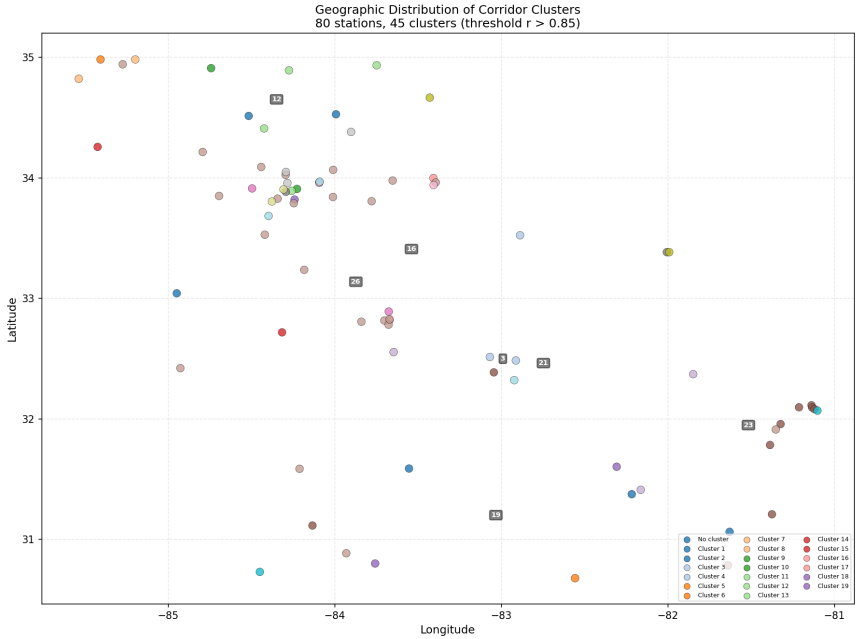






Inter-Station Relationships & Corridor Structure





Pairwise Correlation (top 80 stations)

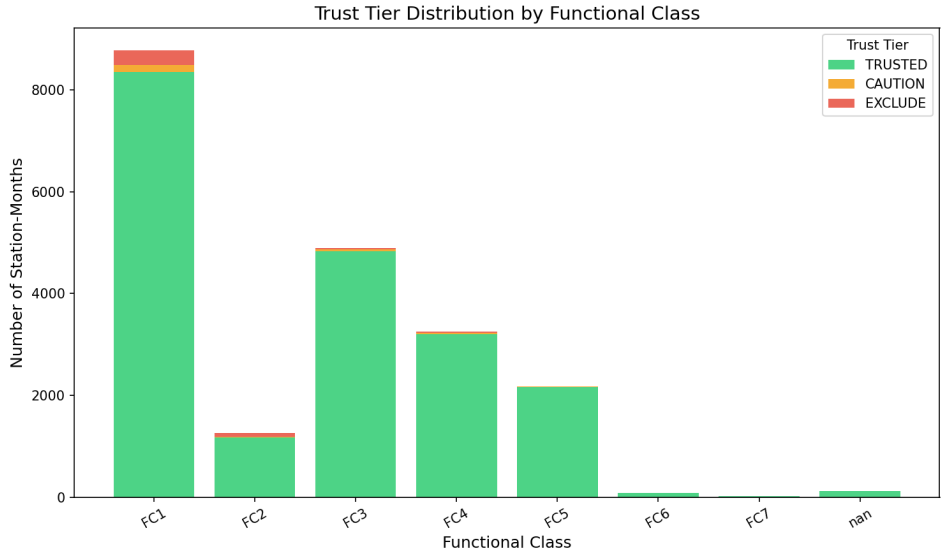
Valid pairs	2,848 (>=180 overl
Mean Pearson r	0.573 (median 0.6
Range	-0.24 to 0.993
High-corr pairs (r>0.85)	410 (14.4%)

Clustering & Reference Stations

- 44 cluster IDs; 7 multi-station corridor clusters
- 37 singletons - no close behavioral neighbors in sample
- 211 behavioral neighbor pairs (r>0.7 AND dist<50km)
- 27 reference station candidates passed all 5 quality criteria
- Top candidate: 089-3323 (DeKalb, Interstate, 10 neighbors)
- Coverage note: top 80 stations only - 150+ eligible not assessed

Anomaly Detection & Station Trustworthiness



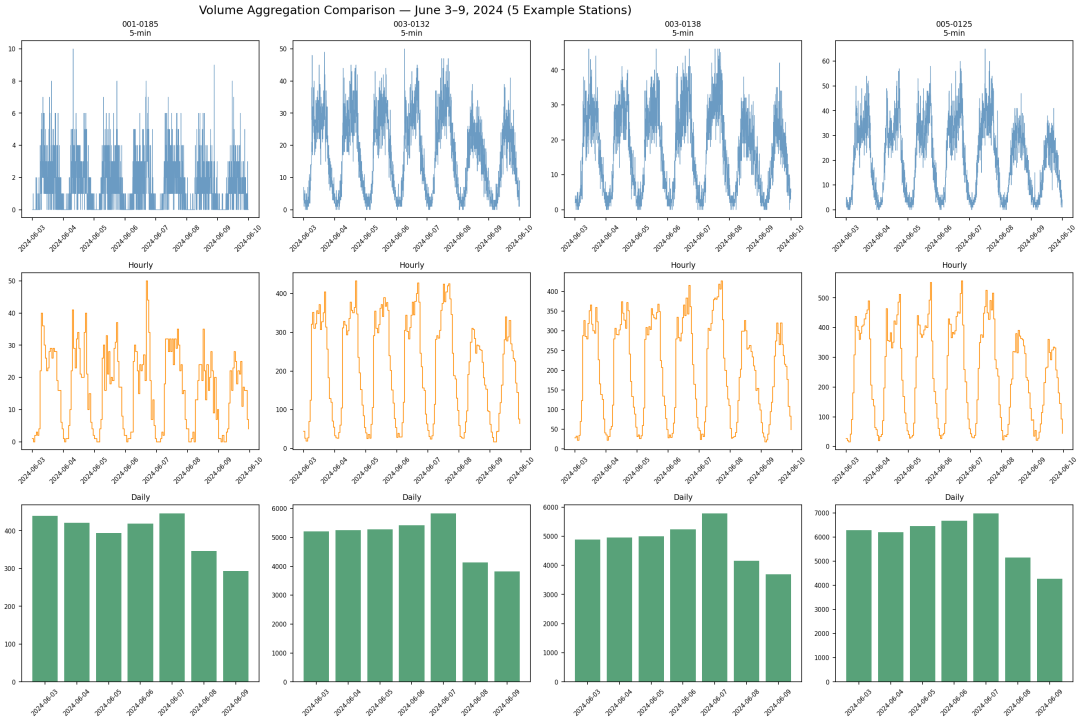


Anomaly Detection (58 stations, 2024)	
Total anomaly-days	391
Explained by event flags	354 (90.5%)
Unexplained anomaly-days	37 (9.5%)

Station-Month Trustworthiness (286 stations)	
TRUSTED months	19,963 (96.9%)
CAUTION months	233 (1.1%)
EXCLUDE months	404 (2.0%)

Station-Level Trust Tiers	
TRUSTED stations	243 (84.9%)
CAUTION stations	36 (12.6%)
EXCLUDED stations	7 (2.5%)

Modeling Readiness - Clean Datasets & Feature Engineering



Clean Hourly Dataset (2024)

Hourly records	818,645
Trusted stations covered	99 of 207 eligible
Temporal span	Full calendar year 2024
Mean coverage	100% (min 50%)
Imputed hours	1.7% (< 6 valid 5-min bins)
Median hourly volume	580 veh/hr (max 17,822)

Station Feature Table

Dimensions	286 stations x 27 features
AADT proxy	median_daily x 365 (June-ref)
Seasonal amplitude	per station from 2023
Trust tier flag	is_trusted_2024
Recommended resolution	Hourly (balances noise vs. signal)

Recommended Modeling Phases

- Phase A: AADT estimation (gradient boosting, 286-station feature table)
- Phase B: Gap-filling via 27 reference station behavioral neighbors
- Phase C: Anomaly-adjusted counts (exclude 37 unexplained days)
- Phase D: Archetype modeling - DEFERRED (needs multi-year data)

Known Gaps & Recommended Next Steps

Open Items (action required before full modeling)

- 100-station pipeline cap: 207 trusted stations eligible; only 99 processed - hourly dataset covers 48% of population. Remove cap and re-run.
- Archetype labels: k-means collapsed (77/78 'Mixed'). Deferred until multi-year directional features and full 207-station sample are available.
- Corridor analysis scope: correlation matrix covers only top 80 of ~230 eligible stations. Expand to full set, relax min_periods 180->90 days.
- 5 chronic-sparse stations (<40% complete months): status unknown - decommissioned, relocated, or persistently failing? Cross-reference GDOT field logs.

Risks & Mitigations

- AADT proxy accuracy: degrades for stations with seasonal amplitude >0.4 - apply seasonal adjustment for high-amplitude stations.
- 2022-23 missingness spike (12%) may bias longitudinal models - flag period; run models with and without.
- Interstate/freeway missingness (13-15%) may reflect multi-lane expected-bin methodology, not sensor failure - validate before imputing.
- 37 unexplained anomaly-days: exclude from model training; audit against weather and incident records.
- Non-standard IDs (021-w334, 185-w227): build explicit COSIT lookup and add station_type flag.

Bottom Line: 243 trusted stations, 818K clean hourly records, and 286-station feature table are ready for AADT modeling.

Holiday effects now quantified. Archetype modeling and full corridor analysis are the two main deferred items.